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Experiment 13: Building Approval-Based Workflow Automation in a Zoho Creator Application

1. Aim

To automate the approval of student project proposals in the Student Project Tracker, so that each proposal moves through a defined review path without manual follow-up, and to keep a clear record of every decision.

2. Theory

Workflow automation replaces manual, paper-based routing of work with rules that run automatically when a record changes. A typical approval workflow has a trigger, such as a new submission; a set of states, such as pending, approved or rejected; a responsible person for each state; and actions, such as notifications and status updates. Automation reduces delays, makes responsibilities explicit and creates an audit trail of who decided what and when.

3. Workflow Design

Stage	Action	Responsible	Next Stage
Submitted	Student submits a project proposal	Student	Pending Review
Pending Review	Faculty guide reviews the proposal	Faculty guide	Approved or Rejected
Approved	Status updated; student notified by email	System	Ongoing
Rejected	Reason recorded; student asked to resubmit	System	Submitted

4. Procedure

- A field named Approval Status with the options Pending Review, Approved and Rejected was added to the Project Entry form.
- A workflow rule was created so that when a new record is submitted, its Approval Status is set to Pending Review and the faculty guide is notified by email.
- A Remarks field was added so that the faculty guide can record the reason for a rejection.
- Approve and Reject buttons were added to the record view, visible only to faculty users, and each button updates the Approval Status field.
- A second workflow rule was created to send an email to the student whenever the Approval Status changes.
- The workflow was tested with three sample records: one approved, one rejected and resubmitted, and one left pending.

5. Test Results

Test Scenario	Expected Outcome	Result
Submit a new proposal	Status becomes Pending Review; guide notified	Pass
Faculty approves a proposal	Status Approved; student emailed	Pass
Faculty rejects with remarks	Status Rejected; remarks visible to student	Pass
Student resubmits rejected proposal	Status returns to Pending Review	Pass
Student tries to press Approve	Button not visible to students	Pass

6. Observations

Every submission moved automatically into the review queue, faculty received notifications, and students were informed of each decision. Rejected records returned to the student with the reason recorded in the Remarks field, and resubmitted records restarted the cycle. No manual reminders were needed during testing. Role-based visibility of the approval buttons prevented students from approving their own proposals.

7. Discussion

The automation improved transparency. Students no longer needed to ask about the status of a proposal, because the status was always visible and each change triggered a notification. The audit trail, which keeps the status history and remarks, supports accountability if a decision is questioned. The design has limitations: a proposal waiting for a faculty member on leave would remain pending, so a reassignment or escalation rule would be needed in a real deployment. Email notifications also depend on correct addresses being stored in the records.

8. Result and Conclusion

The approval process was automated using workflow rules, a status field, remarks and role-based buttons. The automation shortened review time, made the process transparent to all parties and created a traceable record of every decision. The same pattern can be reused for other approval processes such as leave requests or fee waivers.